

Signal Integrity

• eBook •

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CIRCUITS

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In a printed circuit board, a signal propagates through various copper features and components to perform a designated function. If this signal gets distorted or attenuated, it will cause functionality issues in the PCB. In addition, when multiple signals propagate through numerous nets, it is bound to face some integrity challenges. Understanding these problems and mitigating them to achieve an acceptable signal quality is essential to build a quality circuit board.

To have a good signal integrity in a PCB, there are several factors to consider such as stack up, material selection, trace routing, and component placement. Even though it may seem like a gigantic task to maintain good signal quality, particularly in high-density and high-speed circuits, you can avoid most issues just by implementing some basic design principles.

1. What is signal integrity?

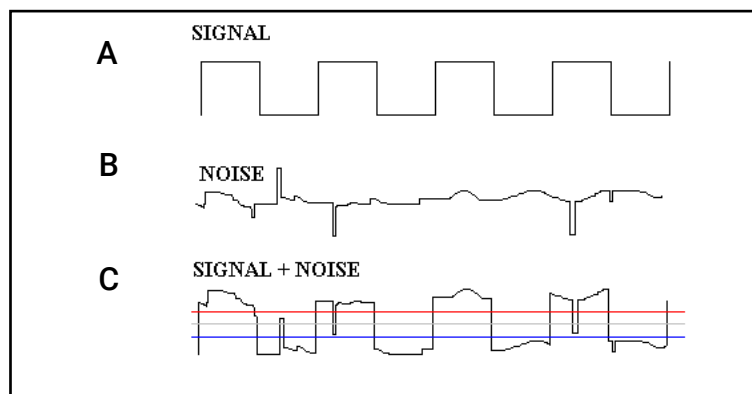
Signal Integrity signifies the signal's ability to propagate without distortion.

Fundamentally, signal integrity issues must be taken care of during the design phase.

To be more descriptive, signal integrity is the measurement of the quality of an electrical signal in a printed circuit board. In digital electronics, a stream of binary values is represented by a voltage (or current) waveform. However, digital signals are fundamentally analog in nature, and all signals are subject to effects such as noise, distortion, and loss.

Over short distances and at low bit rates, a simple transmitting line can transmit with sufficient fidelity. At high bit rates and over longer distances, transmitting lines can have different effects and degrade the electrical signal to the point where errors occur and the system fails.

However, as speed increases, high-frequency effects take over and even the shortest lines can suffer from problems such as ringing, crosstalk, reflections, and ground bounce, seriously hampering the integrity of the signal.



Effect of noise on signal quality

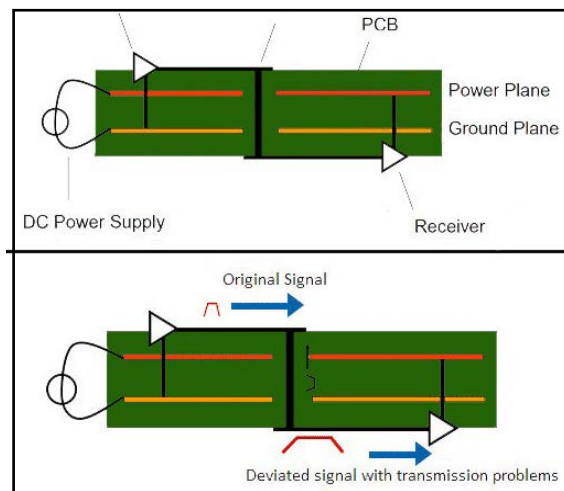
Figure A above shows an ideal signal. Figure C shows the effect of noise on an ideal signal. In the following sections, we discuss the effects of a noisy signal on data integrity. In short, a noisy signal could result in bad data and therefore faulty operations.

2. Need for signal integrity

Signal integrity signifies the signal's ability to propagate without distortion. If there is distortion, then the output of the circuit becomes unreliable. Nowadays, electronic circuit boards are expected to handle high-frequency (from a few GHz to 10 GHz) signals with high-density interconnections. The increased frequency enhances problems such as crosstalk, reflection, EMI, and many more. In these events, the concepts for improving and maintaining signal quality have taken center stage.

A signal is said to have lost its integrity when:

- It gets distorted, i.e. its shape changes from the desired shape.
- Unwanted electrical noise gets superimposed on the signal, degrading its signal to noise (S/N) ratio.
- It creates unwanted noise for other signals and circuits on the board.



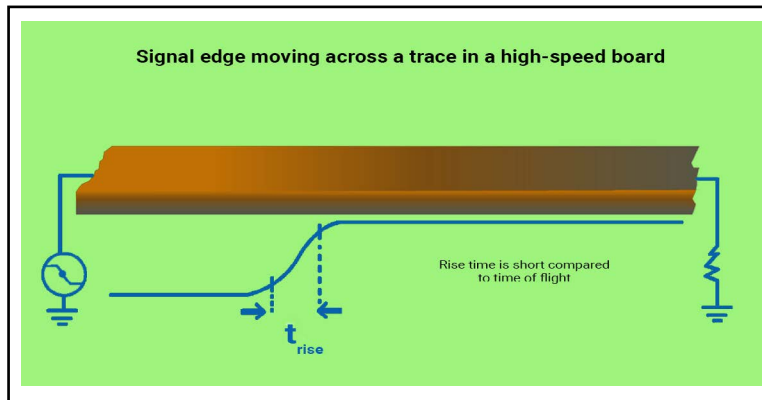
Signal distortion during the propagation

A PCB is said to have requisite signal integrity when:

- All signals within it propagate without distortion.
- Its devices and interconnections are not susceptible to extraneous electrical noise and EMI from other electrical products in its vicinity as per or better than regulatory standards.
- It does not generate or introduce or radiate EMI in other electrical circuits/cables/products either connected to it or present in its vicinity, as per or better than regulatory standards.

3. What leads to signal integrity issues in a PCB?

Perhaps the most important cause of signal integrity issues in a PCB is faster signal rise times. When circuits and devices are operating at low-to-moderate frequencies with moderate rise and fall times, signal integrity problems due to PCB design are rarely an issue. However, when we are operating at high (RF and higher) frequencies, with much shorter signal rise times, signal integrity due to PCB design becomes a very big issue.



Short rise time in a high-speed board

4. Factors that contribute to signal integrity degradation

Fast signal rise times (approximately 5ns) and high-signal frequencies increase signal integrity issues. For analytical purposes, we can divide various signal integrity issues into the following categories:

4.1 Impedance discontinuities in circuit

The conductor needs to have a uniform characteristic impedance throughout its length. If the signal traveling over the trace encounters an impedance discontinuity, it will suffer reflections that cause ringing and signal distortion. A PCB trace can be categorized as a transmission line with the following parameters: resistance, inductance, conductance, and capacitance. Generally

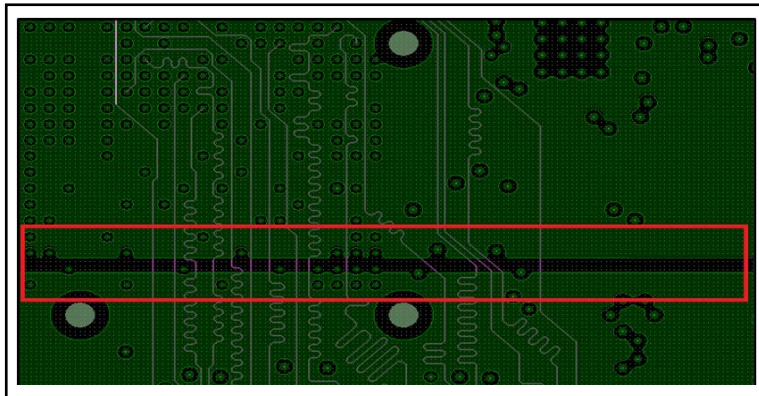
Characteristic impedance of a line = $\sqrt{\text{Inductance} / \text{Capacitance}}$

It is a reasonable assumption as the trace resistance and conductance are negligible at high frequencies. Any factor that alters the ratio between the trace inductance and its capacitance causes impedance discontinuity.

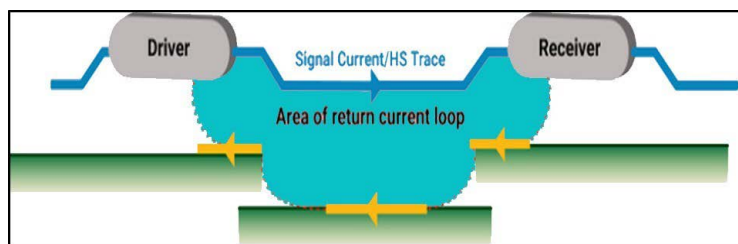
4.1.1 What causes impedance discontinuities?

Discontinuities in the line's impedance will occur in one of the following situations:

- **Impedance variation:** If there is a difference in impedance at the source, destination/load, or the transmission line.
- **Trace impedance variation:** The mutual inductance and mutual capacitance of the line will vary if there is a change in the copper section, geometry, or material of the trace path.
- **Branching of trace:** While interconnecting a circuit, it is quite natural to route a signal to multiple devices. The branches and line stubs utilized for this purpose can modify the impedance value.
- **Splits in return signals:** Typically, a signal with high frequency propagates through a path with the least impedance, which is often the ground layer/return line placed directly under the trace. If there are any physical feature that disrupts this return path, the signal is forced to deviate.

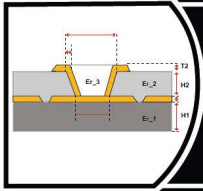


Impedance discontinuity in circuit design



Impedance discontinuity due to split plane

- **Via placement:** Even though they are essential for interconnections, the via size and shape can also modify the trace's inductance and capacitance. Vias also have their own impedance and considering this while designing the circuit is crucial. The top and bottom of the via will introduce two different types of reflection. Normally, this will be in a form of a blip. However, in high-speed design the via impedance discontinuity will be significant.

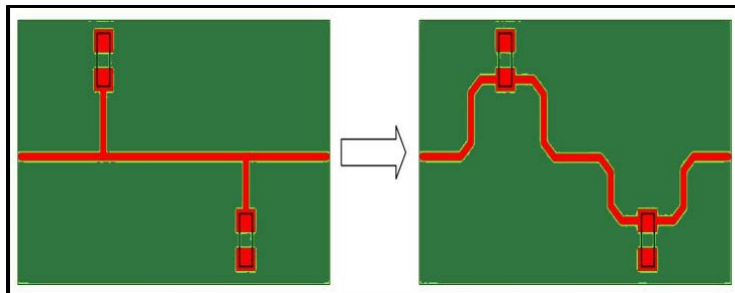


Via Impedance Calculator

Try Now

4.1.2 Mitigating the effects of impedance discontinuity

- You can reduce signal distortion by matching the trace impedance with the source impedance. For this, you can terminate the impedance line using shunt or series resistors at the source and destination.
- Treat all traces as transmission lines and maintain a uniform characteristic impedance and terminate open traces with the right resistance.
- Avoid placing via in the middle of the lines, rather place them close to the source or destination, so you can manage them along with source and destination impedance discontinuities.
- Microvias will help you reduce the length of the stubs. (see section 4.4)
- When a signal line is branching out to several chips, route traces in a daisy chain fashion rather than multi-drop branches. Otherwise, add a matched buffer device to transfer the signal.



Daisy chain trace routing

- Ensure that the return path follows the signal trace. If ground planes are implemented, there should not be any splits interrupting the return path. When you cannot provide a solid ground plane, add a thicker return trace that will cover the whole trace length and is three times the height of the dielectric.

4.1.3 Impedance discontinuity and signal reflection in a transmission line

A signal on a uniform transmission line encounters a constant impedance ' $Z_c = (V/I)$ ' at all locations on the line and the signal travels as desired along it. However, if there is an impedance discontinuity at any point, the signal travel will be affected and signal reflections will occur just like light is reflected when it encounters a discontinuity in the medium through which it travels. Let's say that from point A to point B, the line is uniform with impedance ' Z_i '; but at B, there is an impedance discontinuity, and the line impedance changes at B and beyond to ' Z_o '.

Let's say that a signal is sent from A on the line (of impedance ' Z_i ') towards B. At B, let the incident signal voltage and current be ' V_i ' and ' I_i '. Then, we must have:

$$V_i = Z_i I_i \quad (1)$$

Now, if we look at B from the right-hand side where the line impedance is ' Z_o ', then the transmitted voltage ' V_o ' and the transmitted current ' I_o ' at this point B must be related by:

$$V_o = Z_o I_o \quad (2)$$

It needs to be emphasized that relations (1) and (2) are fundamental to the concept of impedance and must always be obeyed.

Given the values of ' V_i ', ' Z_i ', and ' Z_o ', the problem is to find the value of ' V_o '.

Case 1: $Z_o = Z_i$, (i.e. no discontinuity in the impedance of the line)

In this case, both the above equations (1) and (2) will be simultaneously satisfied by:

$$V_o = V_i \text{ and } I_o = I_i$$

This is to be expected, as in this case, point B is basically any point on a uniform transmission line so that at every point B on this line, the transmitted signal will equal the incident signal.

Case 2: $Z_o \neq Z_i$, (i.e. there is an impedance discontinuity in the line)

Here, by looking at both the equations (1) and (2), since $Z_o \neq Z_i$, it is easy to conclude that we cannot have both $V_i = V_o$ and $I_i = I_o$; because, if for a moment we assume that $V_o = V_i$, then $I_i \neq I_o$; and if we assume that $I_i = I_o$, then $V_o \neq V_i$. Thus we reach the conclusion that at an impedance discontinuity, the incident signal is not fully and completely transmitted onwards.

The only way we can solve this problem satisfactorily is by assuming that a part of the incident signal is reflected back at the point B of impedance discontinuity, and this reflected signal is superimposed on the incident signal at point B and on the line to the left of it; and at point B, the result of the incident signal and the superimposed reflected signal will equal the transmitted signal at point B.

Let's, therefore, say that a part of (V_i, I_i) is reflected back onto the ' Z_i ' transmission line at point B. Let's represent the reflected signal's voltage and current at point B by (V_r, I_r) . Therefore, at point B, the resulting signal voltage due to incident voltage V_i and the reflected voltage V_r would be $(V_i + V_r)$, and as per Kirchhoff Law applied at point B, this must equal ' V_o ':

$$V_o = V_i + V_r \quad (3)$$

And the resulting current due to the superposition of the incident current 'I_i' and the reflected current 'I_r' (which is in a direction opposite to that of the incident current) would be the current 'I₀'. And the reflected (V_r, I_r) signal is traveling on the line with impedance 'Z_i', we must have:

$$V_r = Z_i I_r \quad (5)$$

Using equations (1), (2), and (5) in equation (3), we have:

$$Z_0 I_0 = Z_i I_i + Z_i I_r = (I_i + I_r) Z_i$$

And since, as per (4), $I_0 = (I_i - I_r)$, we have:

$$Z_0 (I_i - I_r) = Z_i (I_i + I_r)$$

Or $I_i (Z_0 - Z_i) = I_r (Z_0 + Z_i)$

V_r/V_i or I_r/I_i indicates the fraction of the incident signal that is reflected back at point B of impedance discontinuity and is called the reflection coefficient 'R_c':

In terms of 'R_c', the transmitted signal voltage and current at point B are given by:

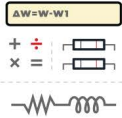
$$V_0 = V_i + V_r = (1 + R_c) \times V_i \quad (7)$$

And: $I_0 = I_i - I_r = (1 - R_c) \times I_i \quad (8)$

From the above it is clear that the value of the reflection coefficient 'R_c' depends on the relative values of 'Z_i' and 'Z₀'; greater the difference between them, more is the amount of reflection. It can be safely concluded that, in case of impedances being purely resistive, the range of values of 'R_c' can be given by:

$$-1 \leq R_c \leq 1$$

And if $Z_0 > Z_i$, a partial positive signal is reflected back leading to a voltage overshoot at B in a pulse signal. And if $Z_0 < Z_i$, a partial negative signal is reflected back leading to a voltage undershoot at B in a pulse signal.



ΔW=W-W1

Impedance Calculator

[Try Now](#)

4.2 Signal reflection, ringing, overshoot, and undershoot

4.2.1 What is reflection?

When a signal is transmitted in a transmission line, some of the signal power may be reflected back to its transmitter rather than being carried all the way along the trace to the far end. Whenever the impedance changes in a circuit, some amount of reflection will happen. The reflected signal will travel back until it encounters another change in impedance and gets reflected again.

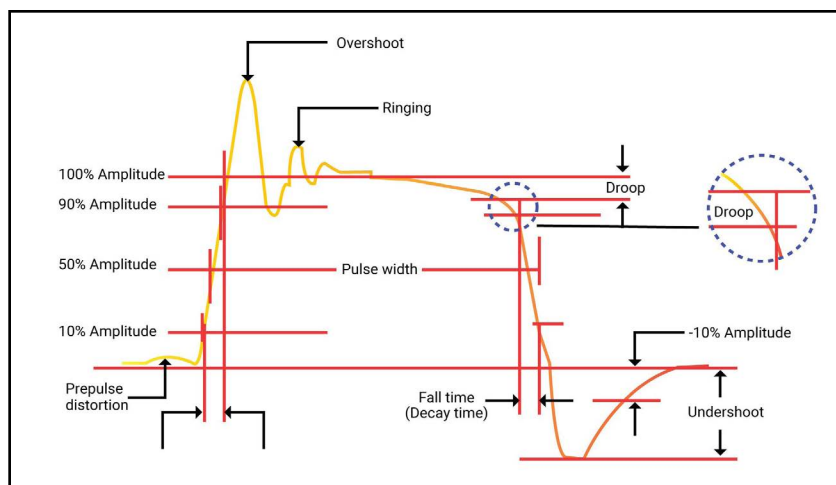
Influence of reflection:

- Signal distortion caused by reflection
- Overshooting and undershooting caused by reflection

How to reduce reflection noise:

- Maintain the constant impedance
- Maintain good ground grading
- Use a series termination resistor and place it near the source point

A theoretical instantaneous transition of the signal allowed maximum upper and lower amplitude.



Signal overshoot and undershoot

1) Overshoot

With reference to the image above, when the signal transits from a lower value to a higher value and the value of the transit signal is more than the actual value, then overshoots occur.

2) Undershoot

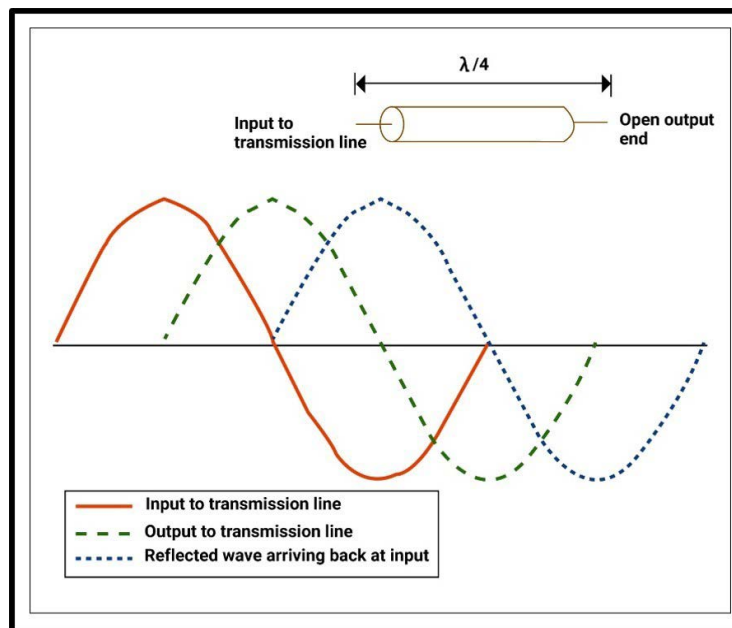
When the signal transits from a higher value to a lower value and the value of the transit signal is less than the actual value, then undershoots occur.

You can implement clamping diodes to limit the peaks of overshoot and undershoot. However, excessive ringing can heat up the diode causing it to burn out. Proper termination of the transmission line is essential to avoid ringing, overshooting, and undershooting. (see section 5.3)

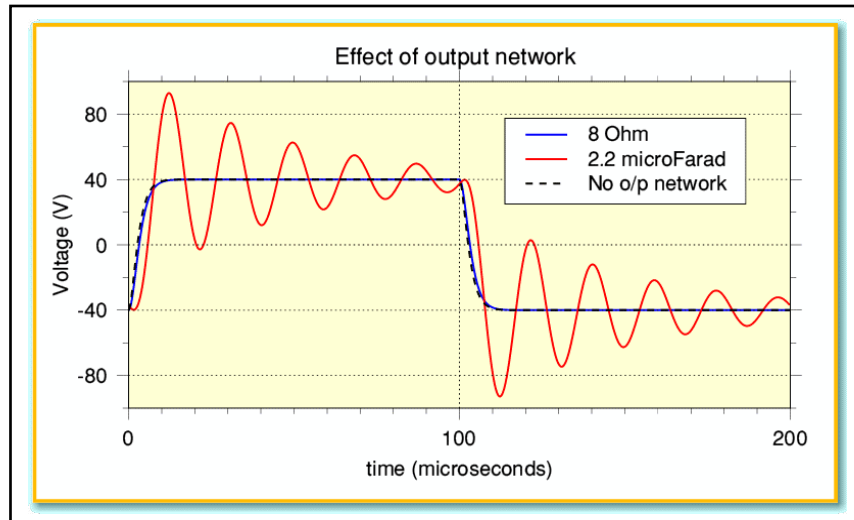
4.2.2 What is ringing?

Ringing is a voltage or current output that oscillates like a ripple on a pond when it's seen on an oscilloscope. The oscillation is a response to a sudden change in the input signal, like turning it on or switching.

Daniel Beeker and **Rick Hartley** explained, "Ringing is the result of having the driver farther away from the receiver than $1/4$ th wavelength. This results in a first-order reflection of more than the incident wave that returns to the driver and becomes a depletion wave at a lower voltage going back to the receiver, until all of the energy finally either goes into the receiver, is converted to heat in the conductors and dielectric or mostly radiates."



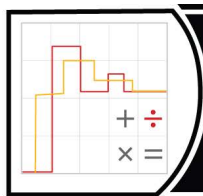
Reflection caused by $1/4$ th the wavelength of the switching speed of the driver



Ringing effects on output network

Influence of ringing:

- Increased EMI
- Increased current flow
- Decreased performance
- Audible feedback

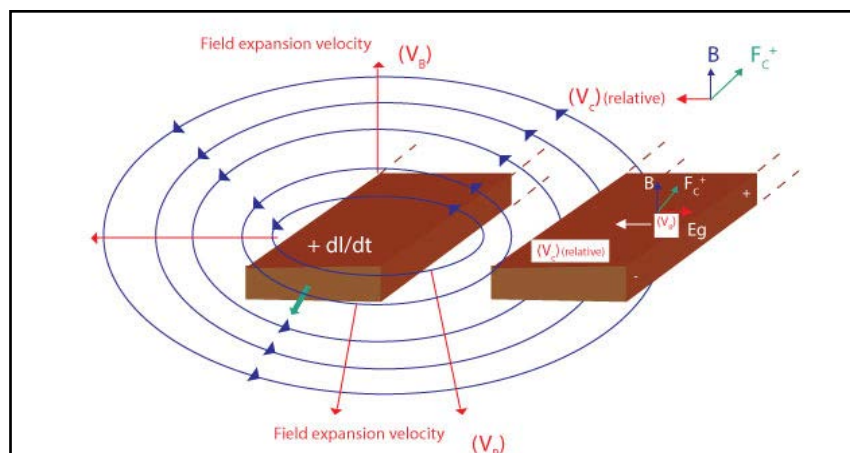


Transmission Line Reflection Calculator

Try Now

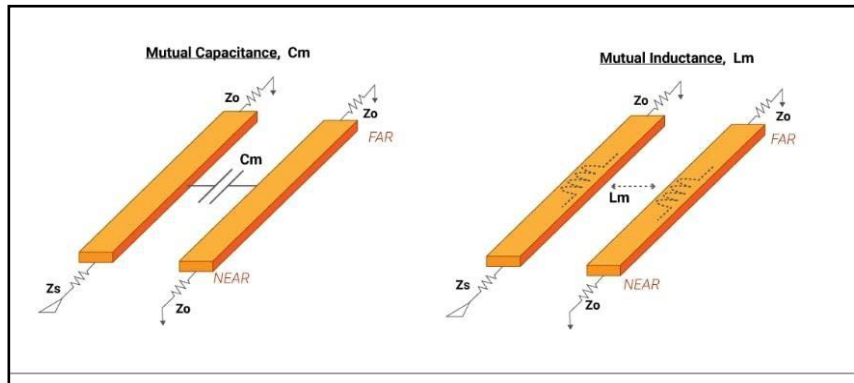
4.3 Crosstalk between two conductors/systems

Crosstalk occurs when a signal propagating in a channel creates an undesired effect on another signal in the neighboring channel (circuit), as shown in the below image.



Crosstalk between two conductors

Crosstalk occurs when there is the coupling of energy from the aggressor signal to the victim signal (typically two tracks close to each other) in terms of the interference of electric and magnetic fields. The electric field is coupled via mutual capacitance between the signals. On the other hand, the magnetic field is coupled via mutual inductance between the signals.



Mutual capacitance and inductance between two conductors

4.3.1 Mutual capacitance between traces

When two traces run parallel to each other and are separated by a dielectric they behave as parallel plates of a capacitor. When the traces are at two different voltages an electric field is generated between them. Any variation of voltage in one of the traces will induce a current in the other trace due to the electric field variation. This capacitance between two traces is called mutual capacitance.

4.3.2 Mutual inductance between traces

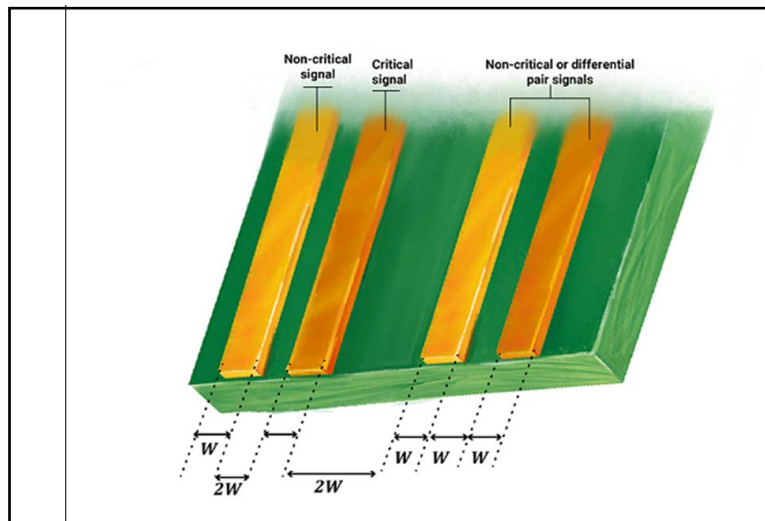
A trace-carrying current has a magnetic field around it. If there is another trace close to the first trace carrying current this magnetic field will couple with the second trace. By Faraday's law if there is a variation in current in the first trace the magnetic field will change which will induce a voltage in the second trace. The inductance due to magnetic field coupled traces is called mutual inductance.

4.3.3 Crosstalk noise in signal line/return path

A fast current transition on a signal/return path may couple onto adjacent signal lines causing a disturbance called crosstalk. The coupling occurs due to mutual capacitance and mutual inductance. In uniform transmission lines, a relative amount of capacitive and inductive coupling is comparable. If there are discontinuities in transmission lines, usually inductive coupling dominates, resulting in switching noise. And as always, faster rise time signals create more crosstalk and switching noise.

4.3.4 Techniques for decreasing crosstalk and switching noise

- Increase the spacing between signal lines as much as routing restrictions allow. The magnitude of the energy in the space is reduced by the square of the distance.
- Implement the 3W rule to lower the crosstalk by 70% and the 10W rule to lower it by 98%.



Optimum spacing between conductors



PCB Conductor Spacing and Voltage Calculator

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- When designing the transmission line, the conductor should be placed as close to the ground plane as possible. This couples the transmission line tightly to the ground plane and decouples it from adjacent signals. Making the signal return paths as wide and uniform as possible. You can implement uniform planes, and avoid split return paths.
- To avoid coupling, the signals should be routed on different layers orthogonal to each other.
- Use a lower dielectric constant PCB material
- Incorporate tightly coupled differential pairs, since they are more immune to crosstalk.
- Reduce parallel run lengths between signals.
- Segregate the group of nets as per their signal amplitude. This will limit the larger voltage (3.3V) nets' influence on smaller voltage nets (1.5V).
- Isolate high-frequency signals and avoid placing asynchronous signals near them.
- Implement guard traces to limit the capacitive crosstalk in transmission lines.

4.3.5 How is crosstalk measured?

Crosstalk is usually determined by the signal percentage that appears on the victim line. It can also be expressed in terms of dB below the driven line level.

The formula for crosstalk is given by:

$$\frac{K (H^2)}{(H^2 + D^2)}$$

Where:

- K is a constant whose value always remains less than 1 and depends upon the rise time of the circuit and the length of the traces experiencing crosstalk.
- H^2 is the product of the height of the parallel traces.
- D^2 is the product of the direct distance between the centerline of the traces.
- The above equation clearly shows that crosstalk can be minimized by reducing H and maximizing D.

Crosstalk in dB is given by:

$$20 \log \frac{V_{victim}}{V_{aggressor}}$$

Where, **V** victim is the voltage on the victim line and **V** aggressor is the voltage on the aggressor line.

4.3.6 What are the different types of crosstalks?

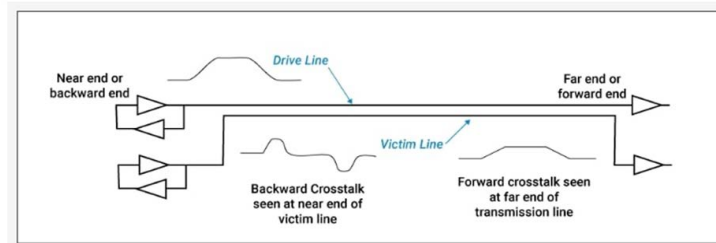
a) Based on the direction of propagation

Forward crosstalk: It propagates along the direction of the aggressor signal.

Forward crosstalk = Capacitive coupling – Inductive coupling

Backward crosstalk: It propagates in the opposite direction of the aggressor signal.

Backward crosstalk = Capacitive coupling + Inductive coupling

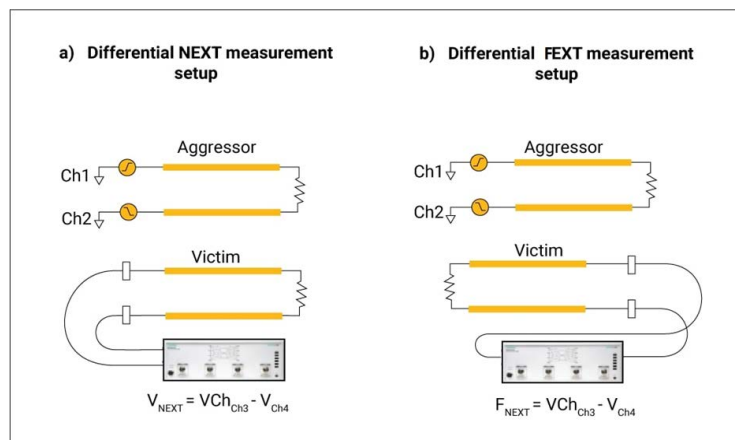


Forward and backward crosstalk

b) Based on the measurement zone

Near-end crosstalk (NEXT): Disturbance on the driver side of the victim line.

Far-end crosstalk (FEXT): Noise on the receiver side of the victim line.



Differential NEXT and FEXT measurement

c) Based on quantification

Power-sum-NEXT(PS-NEXT): Absolute or relative power of near-end crosstalk.

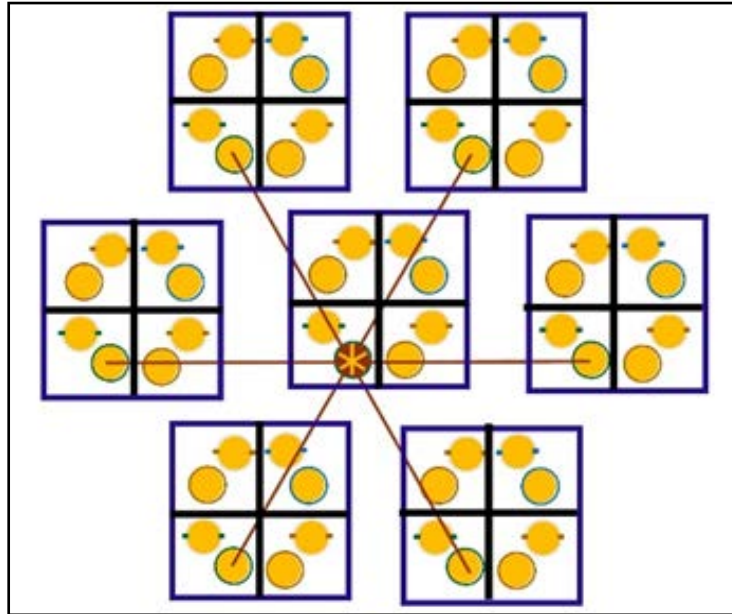
PSNEXT gives total crosstalks from all the adjacent pairs and involves measuring all pair-to-pair groupings relative to power.

Power-sum-FEXT (PS-FEXT): Absolute or relative power of far-end crosstalk.

Power-sum-equal-level-crosstalk (PS-ELFEXT): Sum of PS-NEXT and PS-FEXT.

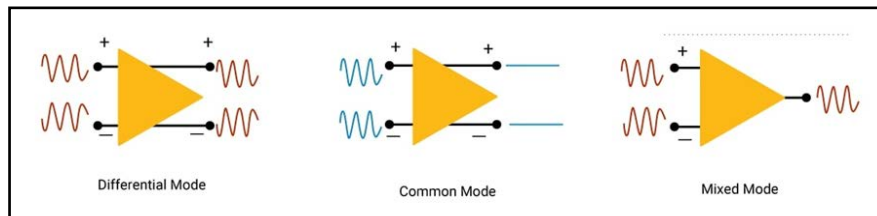
d) Alien crosstalk

Alien crosstalk consists of different frequency content mixed up with the victim signal. The effect becomes so complex that normal strategies such as phase cancellation and shielding are not sufficient.



Alien cross talk

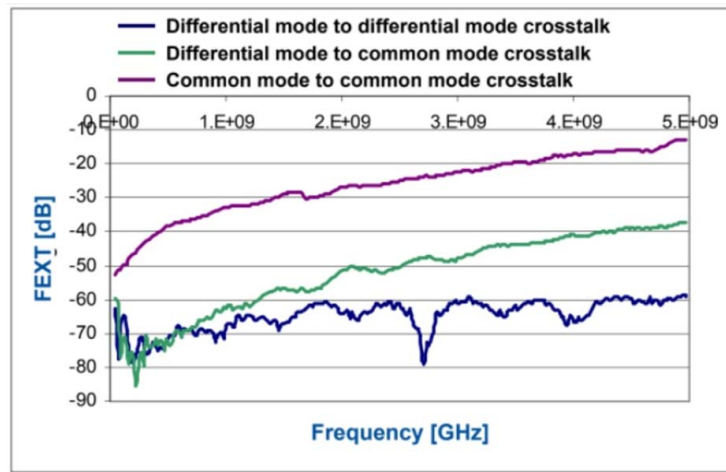
4.3.7 How is crosstalk induced in a differential pair?



Crosstalk in a differential pair

An imbalanced differential system has internal fields that do not cancel completely. The current induced from external fields will be no longer in equal amplitude and opposite in phase, so they do not cancel out either. This will result in a common mode current which has more adverse effects than the differential mode.

To limit the crosstalk, it is essential to match the lengths of the differential pairs within the 5 mil tolerance. Also, the spacing between the pairs, especially at U bends should be kept at $3W$ (W is the width of trace). If there is a phase bump, its height should not exceed twice the spacing ($2s$) between the differential pair.



Comparison between common mode and differential mode crosstalk effects with respect to frequency.

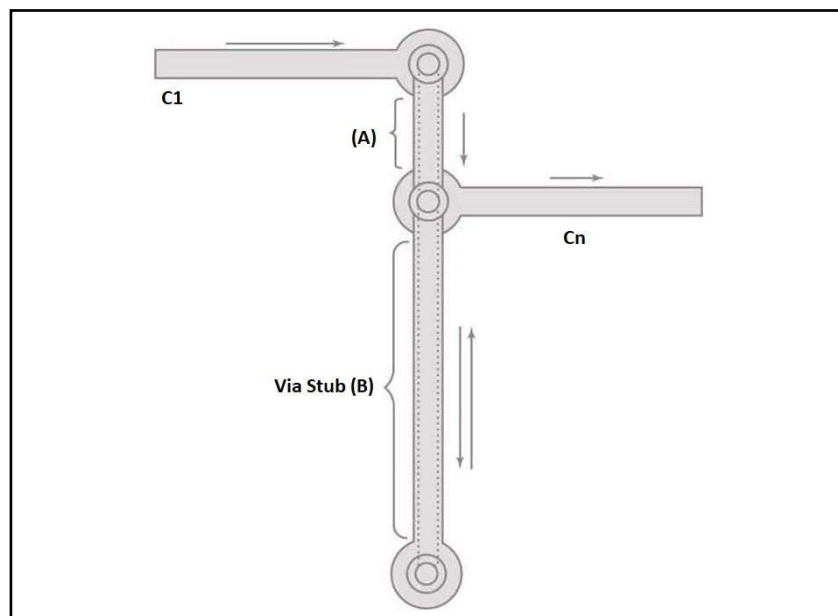
Image credit: Intel

4.4 Via stub

4.4.1 What is a via stub?

When a routed signal starts from the top layer and ends with some inner layer, the remaining portion from the inner layer to the bottom layer is a via stub.

Daniel Beeker and **Rick Hartley** said, "A stub is a single piece of conductor, and unless there is a pair of vias next to each other - one ground and one signal - or a signal via and a ground plane, the field does not see the stub except as very high impedance."



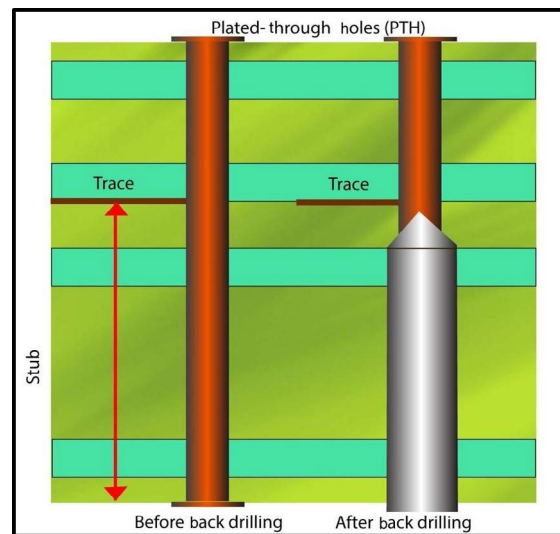
Signal reflection due to via stub

In the above figure, the signal traveling from C1 (outer layer conductor) through part A splits at the junction between via and Cn (inner layer conductor). One part of the signal propagates through the Cn whereas another part travels through via stub (B) which gets reflected at the point of junction. Here, the reflected signal again splits and reaches both the source and the destination through C1 and Cn.

Basically, the via stub acts like a resonant circuit with a specific resonant frequency at which it stores maximum energy within it. If the signal has a significant component operating at/near that resonant frequency, that component of the signal will be heavily attenuated due to the energy demands of the via stub. This effect continues for the fundamental resonant frequency as well as the odd multiples of the fundamental frequency.

4.4.2 How to avoid via stubs?

One of the ways to mitigate this phenomenon is by ensuring that the maximum frequency of the signal is always less than the fundamental resonant frequency of the via stub. You can also adapt a [back drilling technique](#) to eliminate the via stubs by creating a larger diameter hole.



Back drilling to remove via stubs

4.4.3 Why does a via stub act like a resonant circuit?

A signal comprises various sinusoidal components of different frequencies and amplitudes as indicated by its spectrum or Fourier's transform. If ' f ' is the frequency of a sinusoidal component of the signal, then its velocity ' V ' in a medium of dielectric constant ' ϵ_{reff} ' is given by:

$$V = \frac{V_c}{\sqrt{E_{eff} f}}$$

where,

'Vc' = velocity of light in vacuum = $2.99792458 \times 10^{10}$ cm/sec = 11.8028527×10^9 inches/sec

And the time period of one wavelength is:

$$t_{1\lambda} = \frac{1}{f}$$

So that the time for quarter wavelength is:

$$t_{\frac{1}{4}\lambda} = \frac{1}{4} t_{1\lambda} = \frac{1}{4f}$$

The resonant frequency 'frs' of a stub of length 'ls' is that frequency of a sinusoidal wave whose quarter wavelength equals the length 'ls' of the stub.

Now, the time taken by a signal traveling at speed 'V' to travel a length 'ls' is given by:

$$t_{ls} = \frac{ls}{V} = \frac{ls\sqrt{E_{eff} f}}{V_c}$$

At resonant frequency 'frs', as per the definition given above, this time should equal the time taken by a quarter of the wave of frequency 'frs'

$$\frac{1}{4}t_{1\lambda} = \frac{1}{4f_{rs}} = t_{ls} = \frac{ls}{V} = \frac{ls\sqrt{E_{eff}}}{V_c}$$

Which gives us the following expression for 'frs':

$$f_{rs} = \frac{V_c}{4ls\sqrt{E_{eff}}} = \frac{11.8}{4ls\sqrt{E_{eff}}} \text{ GHz}; \quad ls \text{ in inches}$$

A signal of this frequency will take half wavelength time to come back from the stub to the junction point again and will, therefore, be 180 degrees out of phase with the original signal, thereby considerably attenuating the signal at the junction point.

Thus, the stub acts like a 'devourer' for signal energy in a very narrow band around this 'frs' frequency – it will store any incoming energy of frequency 'frs' in itself and not allow it to propagate further. These are precisely the characteristics of a resonant circuit and that is why 'frs' is called the resonant frequency of the stub.

Actually, 'frs' is the fundamental (or lowest) resonant frequency of the stub. For signal integrity, recall that it is the frequency at which the stub length equals one-quarter wavelength and total to and fro propagation time equals 1/2 the time period of one wavelength.

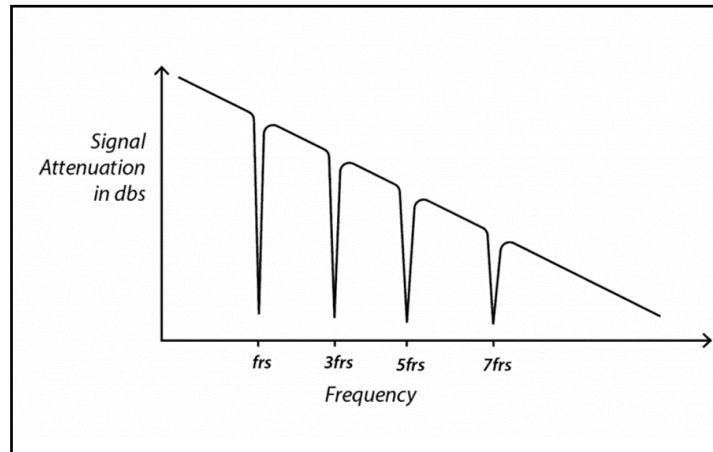
The next resonant frequency 'frs1' will be such that to and fro propagation time equals $(1 + \frac{1}{2} = \frac{3}{2})$ time periods of one wavelength or one direction propagation time over the stub length equals $\frac{3}{4}$ time period of a wavelength.

$$\frac{3}{4}t_{1\lambda} = \frac{3}{4f_{rs1}} = \frac{ls\sqrt{E_{eff}}}{V_c} = \frac{1}{4f_{rs}}$$

From this, we get: $f_{rs1} = 3f_{rs}$

And similarly, $f_{rs2} = 5f_{rs}$... and so on.

Thus, in the plot of the signal attenuation vs frequency, we will see sharp 'nulls' at 'frs', '3frs', '5frs'... frequencies – basically at the fundamental resonant frequency and its odd harmonics as illustrated below:



Signal attenuation due to via stub vs. frequency

4.4.4 Maximum allowable via stub length

If 'fmax' is the maximum frequency component of a signal that needs to be faithfully transmitted, then the resonant frequency of the via stub should exceed 'fmax' by a good enough margin – a 40% or more margin is sufficient. Therefore:

$$f_{rs} = \frac{11.8}{4ls\sqrt{E_{eff}}} \geq R f_{max}; \quad R = 1.4$$

This gives us for 'ls':

$$ls \leq \frac{11.8}{4\sqrt{E_{eff}} R f_{max}} \text{ inch}; \quad f_{max} \text{ in GHz}; \quad R = 1.4$$



**Maximum Via Stub
Length Calculator**

[Try Now](#)

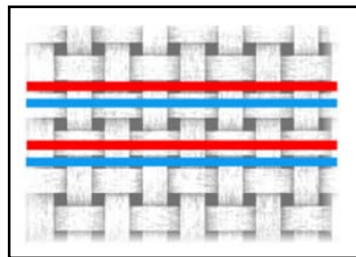
4.5 Skew and Jitter

Signals take finite times as they travel on a PCB from source to receiver. The signal delays are proportional directly to signal line lengths and inversely proportional to signal speed on the specific PCB layers. If data signals and clock signals do not match in overall delays, they would arrive at different times for detection at the receiver, and this would cause signal skews; and excessive skew would cause signal sampling errors. As signal speeds become higher, the sampling rates are also higher, and allowable skew gets smaller, causing a greater propensity for errors due to skew.

Skew in the signal can be caused due to signal integrity issues such as crosstalk, reflection, switching noise, etc. It can also be a result of the fiber weave effect in the dielectric material. Proper grounding and termination of transmission lines is the best method to overcome these challenges. Skew in a group of signal lines can be minimized by signal delay matching, primarily by trace length matching.

4.5.1 Fiber weave effect

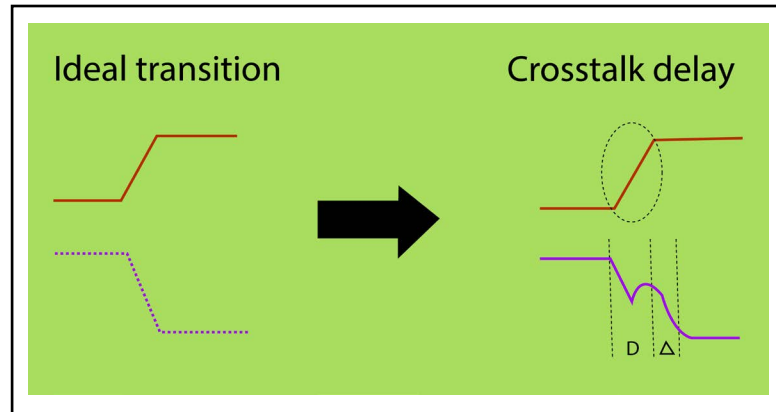
Fundamentally, the glass weave (high Dk) and the resin (low Dk) which makes up the PCB substrates have different dielectric constants.



Fiber weave effect on two signals (red and blue)

In the above image, one of the two signals (red) is traveling only on the glass weave it experiences uniform Dk distribution whereas the other signal (blue) alternates between glass weave and resin. This creates a variation in Dk distribution which causes skew in the propagating signal called the fiber weave effect.

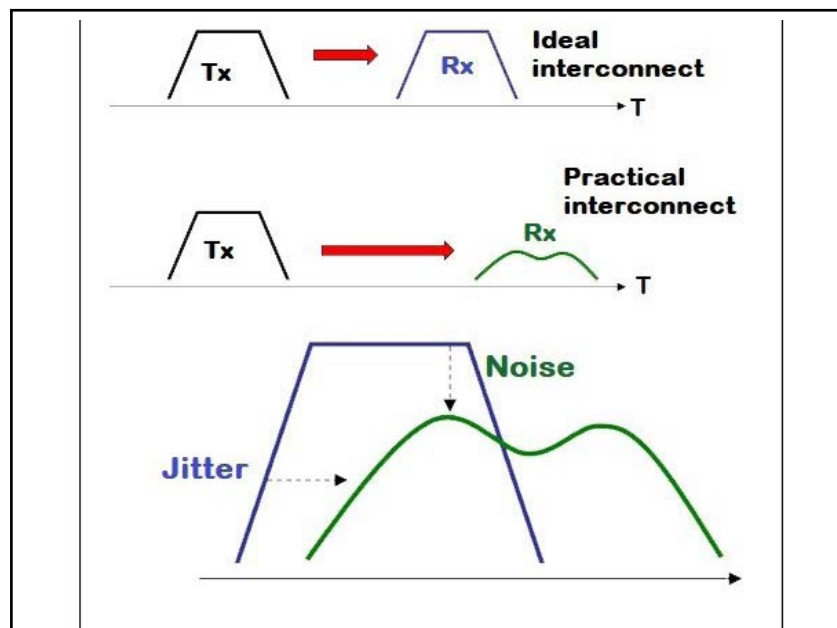
You can choose a tighter glass weave composition to avoid this effect. You can also use zig-zag trace routing to reduce the skewing.



Delay in signal due to crosstalk

4.5.2 Jitter in the signal

Generally, the interconnects not only introduce a delay but also alter the signal's amplitude. This modification to the original signal is called jitter or amplitude noise.



Effects of jitter on the output signal

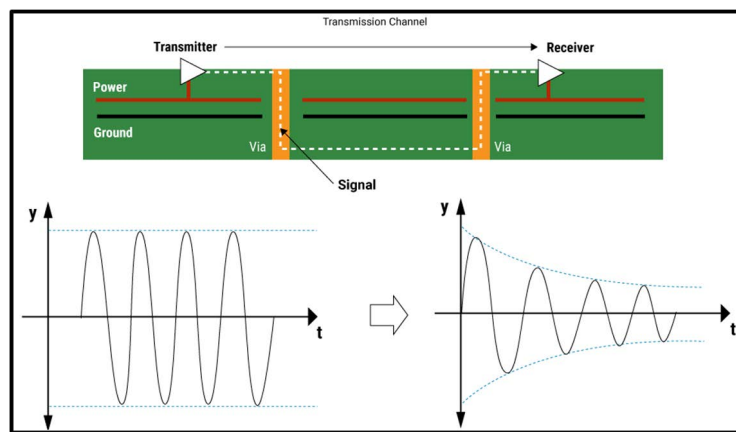
Jitters present in the clock signal affect the performance of digital circuits, especially high-speed.

Minimize impedance or avoid phase discontinuity in impedance at the output and input of the regulator to mitigate clock jitters.

4.6 Signal attenuation

Signals suffer attenuation as they propagate over PCB lines due to losses caused by conducting trace resistances (which increase at higher frequencies due to skin effect) and dielectric material dissipation factor D_f . Both these losses increase as frequency increases, therefore higher frequency components of signals will suffer greater attenuation than do the lower frequency components; this causes a reduction in signal bandwidth, which then leads to signal distortion by an increase in signal rise time; and excessive signal rise time increase results in errors in data detection.

When signal attenuation is an important consideration, one has to choose the right type of low-loss high-speed material and proper control over trace geometries to minimize signal losses.



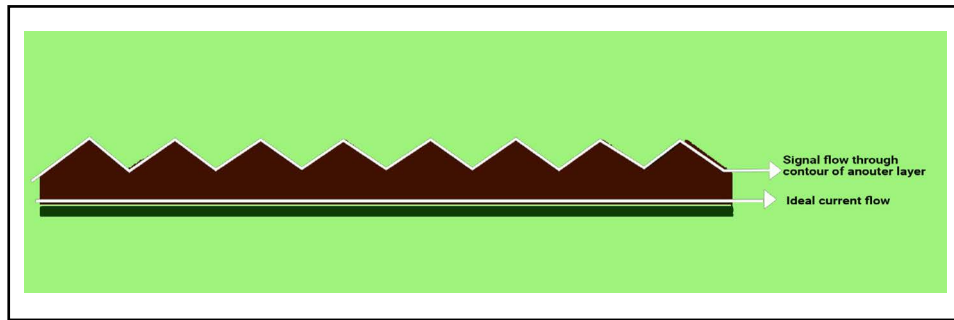
Signal attenuation at the output

4.7 Skin effect

The varying magnetic fields induce an electric field when an alternating current propagates through a trace. The induced electric field opposes the main current flowing through the conductor center but aids the current on the conductor's outer surface. This increase in current density on the conductor surface is called the skin effect.

Skin effect results in copper loss and eventually signal loss. This effect is absent in DC circuits. On the contrary, the impact increases with the increase in frequency of AC circuits. This can be mitigated by implementing wider traces with a larger surface area, reducing the current density.

Another reason for the increased skin effect is the rough/toothy surface of the trace which enhances the effective length and copper loss. Hence, opting for a smoother surface as much as possible may help you regulate the skin effect and improve signal integrity.



Skin effect on the outer surface of a conductor

4.8 Ground bounce

Ground bounce is a form of noise that occurs during transistor switching i.e. when the PCB ground and the die package ground are at different voltages. It interrupts high-speed or high-frequency operations. The major cause for a ground bounce is the difference in ground potentials at various points in a circuit.

During assembly, grounding is provided to all the components and traces. Due to certain discrepancies, there are chances of potential differences between various ground points. This causes ground bounce.

Ideally, for an IC package mounted on board:

Potential difference between the
voltage source pin and ground of
an IC

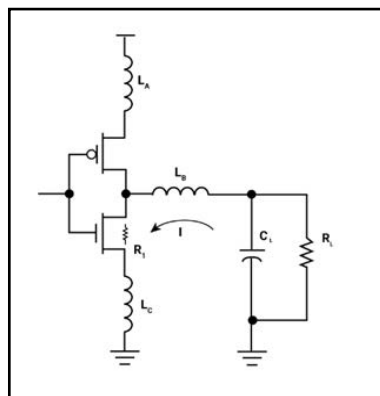
=

Potential difference between the
voltage source pin and ground
of a PCB

However, the potential differences are not always the same.

4.8.1 How is ground bounce generated?

The ground bounce generation can be explained using the sample circuit given below.



A model circuit for illustration of ground bounce

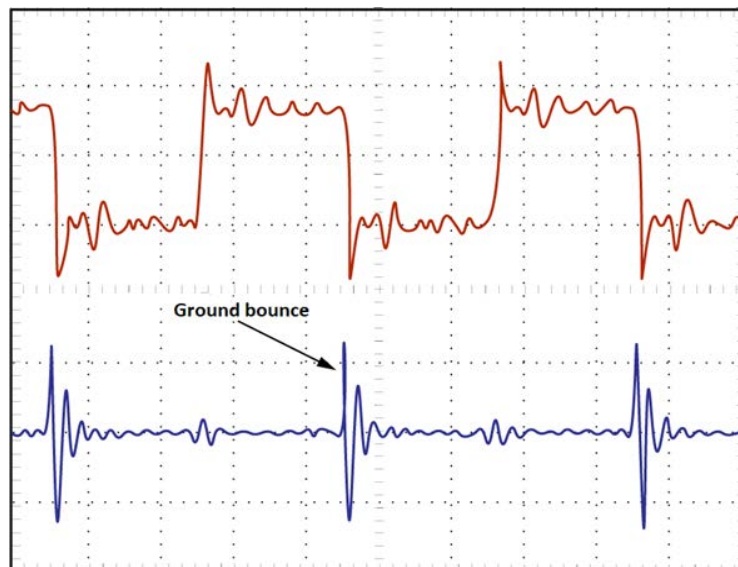
In the above circuit, a CMOS is connected to an output load with CL capacitance and RL resistance. Other components include

- LA: intrinsic inductance in the power lead of the package
- LB: intrinsic inductance in the output lead of the package
- LC: intrinsic inductance in the ground lead of the CMOS package
- RI: intrinsic resistance of the CMOS IC

Consider a scenario when the output varies from high to low. Here, the current flows out of the load side, and load capacitance discharges. The flow of this current (I) across the inductances LB and LC generates a voltage, $V = L(di/dt)$.

The generated voltage modifies the internal CMOS ground to vary from the external load ground. Hence, the internal reference ground is at a higher potential than the external ground's zero potential. This difference generates noise in the circuit and impacts the switching device's functions in the circuit.

The ground bounce oscillations of a circuit observed through an oscilloscope are shown below.

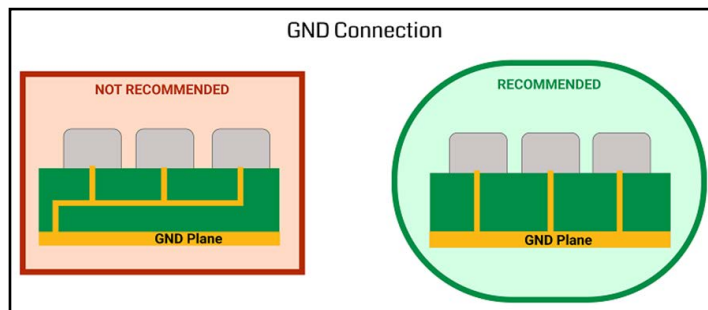


Ground bounce waveforms

The top waveform represents the output of the I/O pins of the switching device. The bottom waveform shows the spikes (noise) due to ground bounce.

4.8.2 Techniques for decreasing ground bounce

- One of the common methods is by implementing a bypass capacitor between the VCC and GND pins of an IC package. It effectively bypasses the voltage spikes and power supply noises.
- Wherever possible try to implement via-in-pads.
- Use a short signal return path as it will reduce the parasitic capacitance.
- Do not use sockets or wire-wrap boards.
- Never share ground vias or traces for ground connections. Use individual vias and traces to connect to the ground plane.



Recommended ground connections

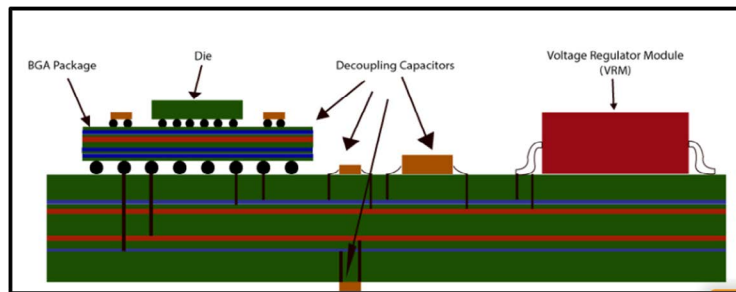
- Use low voltage differential signaling (LVDS) as the I/O standard. This standard offers high bandwidth and high noise immunity.
- Choose packages with short leads to reduce the series inductance. BGAs are also recommended.
- Use solid ground planes to reduce IR losses and inductance. Avoid ground split planes.
- Try to use lower switching components.
- Incorporate serially-connected current-limiting resistors

4.9 Power distribution network noise

Power and ground rails/paths/planes have very low, but finite nonzero impedances. When you switch the states of input signals and internal gates, the currents through the power and ground features (rails/paths/planes) change, causing a voltage drop in power and ground paths. When the occurrence of such instances increases, there will be a more significant voltage drop across power and ground rails. This results in excessive noise in the system that may lead to device malfunction.

To mitigate these effects, you can take the following measures:

- Minimize the impedance of the power system in the PDN.
- Power and ground planes should be placed as close as possible.
- Multiple low-inductance decoupling capacitors can be used across power and ground rails. They should be placed as close to the power and ground pins as possible.



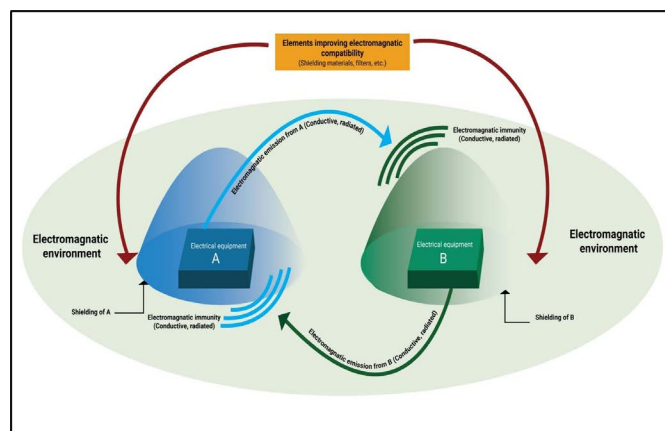
Implementing decoupling capacitors to reduce PDN noise

Try Now

Signal and Plane Layer Estimator

4.10 EMI

An electronic system includes many components such as PCB, ICs, I/O cables, etc. At high frequencies (in GHz) the interconnect can act as an antenna and create EMI which will interfere with other operating components in the vicinity. This depends on the length and the current carried by the interconnect.



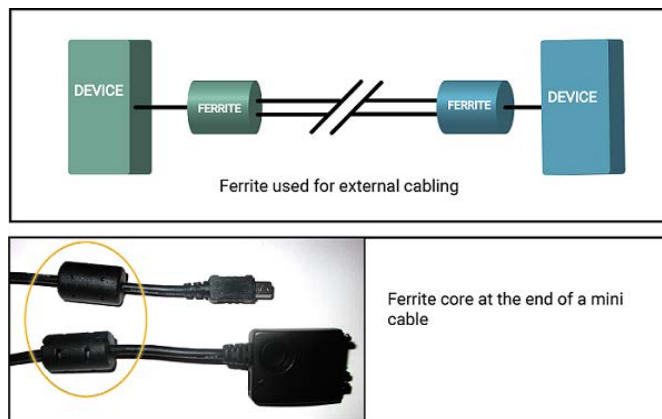
Electromagnetic interference between two systems

4.10.1 Sources of EMI

EMI constitute both common and differential mode radiations.

Two types of EM emission:

a) **Conducted emission:** Penetrates through power input lines and cables. It can be limited by incorporating line filters near power input or connectors. You can also implement a ferrite core/ring/bead that uses high-frequency current dissipation in a ferrite ceramic which acts as a noise suppression device.



Conducted interference can be reduced using a ferrite core/ferrite ring.

b) **Radiated emission:** Occurs due to switching devices, electrostatic discharges, and EM waves in power/communication lines. It can travel through the air from electronic devices and traces, to interfere with other electronic systems.

Other EMI sources:

- High-frequency traces
- Common-mode (CM) currents
- Differential-mode (DM) currents
- Poor decoupling practices at power and ground planes result in unintentional common-mode (CM) and differential-mode (DM) currents.

4.10.2 Principle of EMI reduction

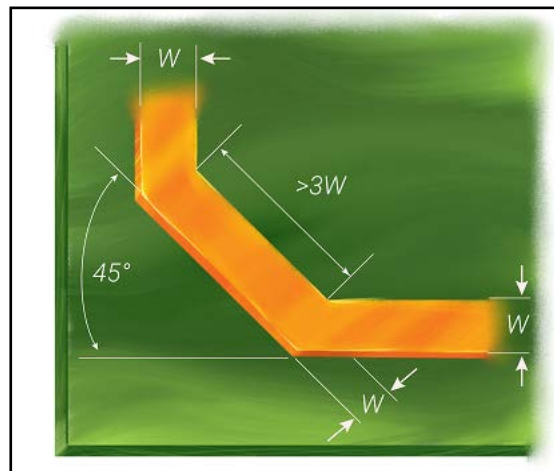
Faraday's law states that **the magnetic field generated by a coil is directly proportional to the area of the coil and the current.**

$$E = di/dt \propto \text{Coil Area} \propto \text{Current going through the coil}$$

If the **coil/loop area reduces**, the magnetic flux will also reduce. We can achieve this by implementing a solid ground plane close to the signal plane. The minimum dielectric thickness of the single layer is around 3 mils (3000th part of an inch), and thus the area becomes very small. We can also minimize the rate at which the current changes and impedance mismatches.

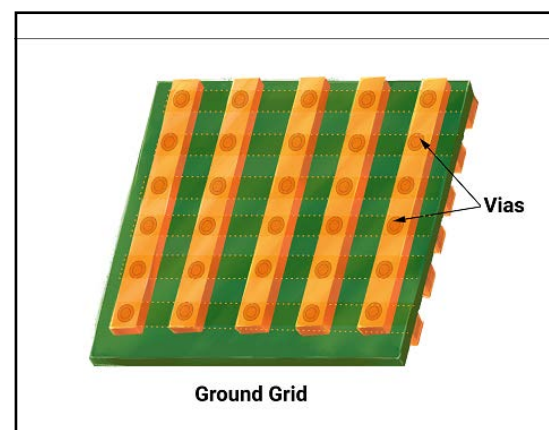
4.10.3 Techniques for reducing EMI

- Reduce the length and area of the signal return paths.
- Route traces from components to the processor by avoiding any void in return path.
- Implement surface-mount devices (SMD) instead of leaded devices. SMD offers lower inductances in comparison with leaded components. Avoid implementing a large number of through-hole components.
- 90° bend trace can act as an antenna and can impact capacitance and characteristic impedance resulting in reflection. To avoid this, implement 45° turns.



Implementing 45° bend in traces

- Stick to the 3W rule for separation between adjacent traces.
- Run proper ground plane. If you cannot provide a solid ground plane then implement ground grids.



Implementing ground grids on PCB

- Signal vias have their own capacitance and inductance effect, so avoid placing them on the same layer as critical traces to avoid impedance mismatch. If you have to implement a via on trace then connect a ground via along with it.
- For clock circuits, do not forget to add decoupling capacitors. Guard and shunt traces are also implemented to protect clock lines from EMI sources.

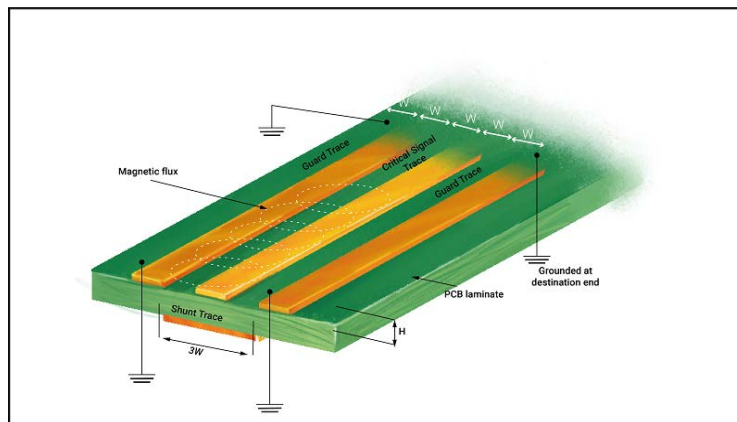
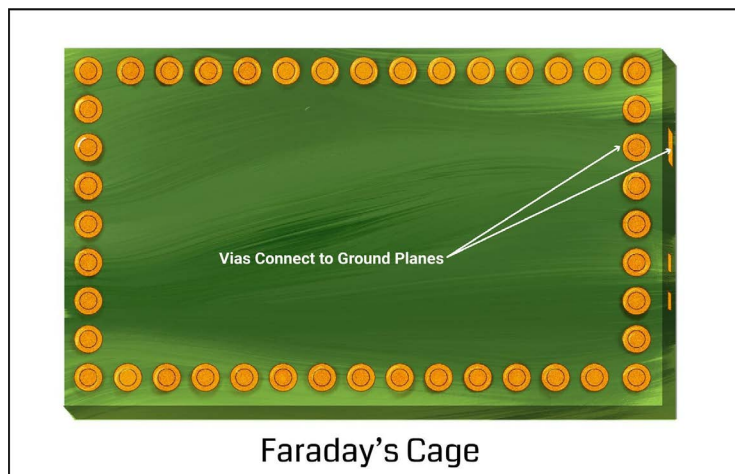


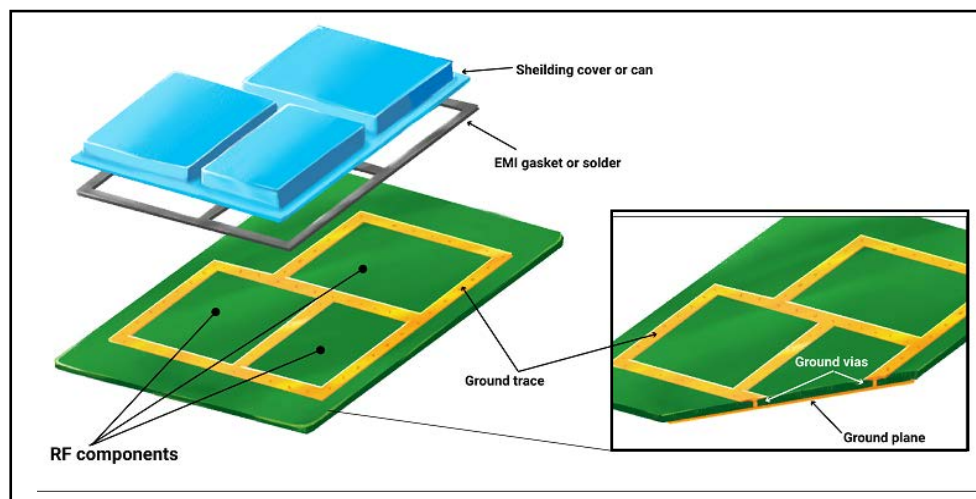
Illustration of Guard and shunt traces

- Create a Faraday cage by incorporating ground across the edge of the circuit board. This will limit the EMI within the defined boundary.



Faraday cage on circuit board outline

- The copper fills should be grounded properly to avoid floating copper (acts as an antenna).
- Incorporate EMI shielding by covering the whole system or a part of it in a conductive/magnetic material connected to the ground. This reduces the size of loop antennas by absorbing and reflecting a part of their radiation. EMI/EMC shielding can also protect the signals from external noise.



Shielding to mitigate EMI

- Group and segregate signals such as analog, digital, power supply, low-speed, high-speed signals, etc. Traces for each component should be routed in the defined area. Use filters when different types of signals are communicating with each other.

5. Best design practices to lower signal integrity issues

To build an efficient and error-free PCB, it is essential to consider signal integrity issues from the design phase. Incorporating design changes from an early stage will go a long way in avoiding or even eliminating signal integrity issues. The goal is to ensure that the signal traveling from the input to the output should not be degraded by any interference.

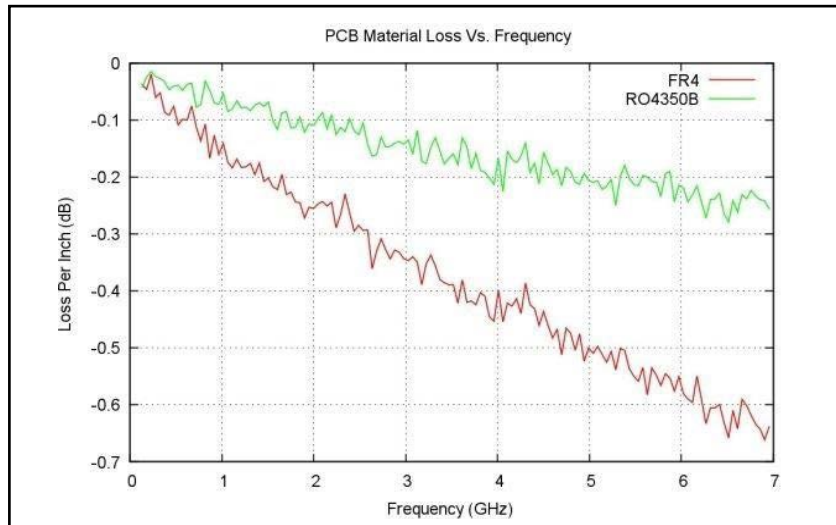
5.1 Material selection

The electrical properties of a substrate greatly affect the signal integrity of the circuit. When the molecules of the dielectric medium get excited at high frequencies, they absorb some of the signal energy leading to signal loss. This phenomenon is known as dielectric absorption (loss factor). The signal loss and attenuation caused while propagating through the substrate will be substantial if you have not selected the right material.

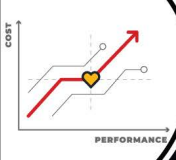
The dielectric constant (D_k) is also one of the determining factors for maintaining uniform impedance of the circuit to avoid crosstalk and noise. If you are planning to include a targeted controlled impedance, it is important to consider the D_k of the material along with the trace width. Utilizing the D_k , you can determine the velocity (V_p) and the propagation delay (t_{PD}) of the signal. These factors can provide insight into the length at which a signal trace should be considered a transmission line.

Nowadays, FR4 is prevalently used in PCB designs due to its excellent electrical properties and affordability. They are a good choice for frequencies ranging below 2.5-3 GHz. For frequencies above this range, high-speed laminates such as Rogers will be a better choice. This is because for frequencies beyond 5GHz the D_k of FR4 reduces. In contrast, the dielectric constant of Roger material (R04350) remains constant from 0 Hz to 15 GHz. Hence, the selection of reliable material is vital to maintain good signal quality.

The graph below represents the insertion loss per inch (energy lost by the signal propagating through a cable) increasing with the frequency for FR4 and Rogers R04350B material. If the insertion loss is higher, then signal attenuation will also be higher.



Loss vs. frequency of FR4 and Rogers RO4350B material

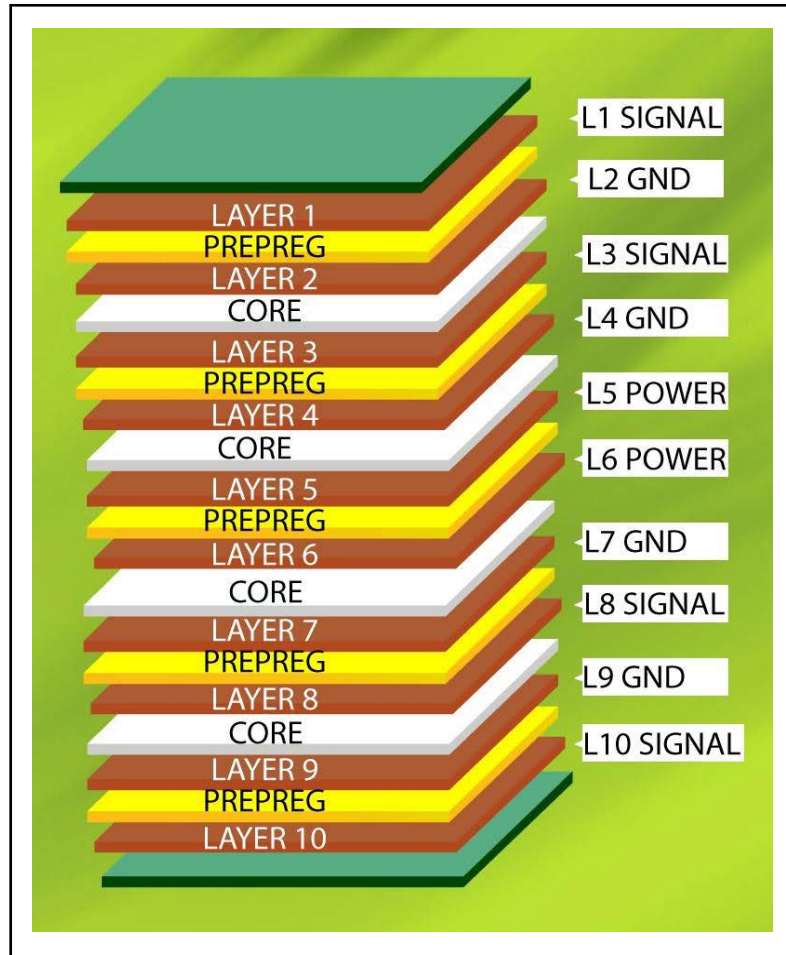


PCB Material Selector

[Try Now](#)

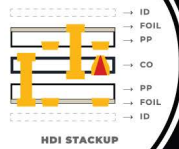
5.2 Stack-up preparation

As discussed in the previous sections, many signal integrity issues are rooted in the unavailability of a proper return path for the signal. An appropriate stack-up design can help you mitigate this problem. To provide obstruction-free and short return paths, you need to place the reference or ground layer adjacent to the signal layer. Especially for sensitive or controlled impedance routes, it is essential to place the reference plane as close as possible. If the stack-up is built properly, you can also achieve better power integrity along with signal integrity.



Typical 10-layer PCB stack-up

The image shows a typical 10-layer stack-up construction. Generally, an alternating signal and ground plane are built to provide shielding from reflections and EMI issues. This also determines the microstrip and stripline configurations that impact losses in the digital and analog signals. A microstrip has a single reference plane whereas in a stripline configuration, the signal layer is sandwiched between two ground layers.



PCB Stackup Designer

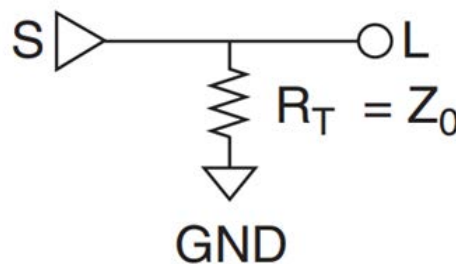
Try Now

5.3 Trace termination techniques

Many signal integrity issues can be resolved by implementing a proper termination technique. There are six different ways to terminate a transmission line. Select a suitable scheme based on your design requirements.

5.3.1 Parallel termination

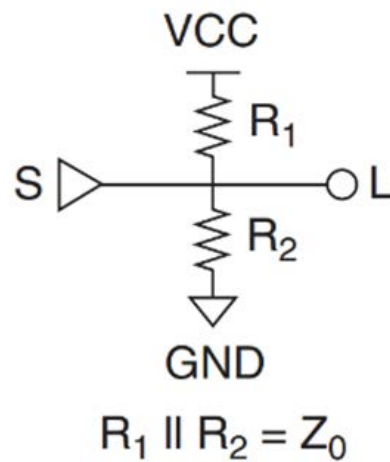
Here, the termination resistor (R_T) is placed nearer to the load of the circuit to achieve maximum efficiency at a high-output state. The resistor value should be equal to the line impedance.



Parallel termination

5.3.2 Thevenin termination

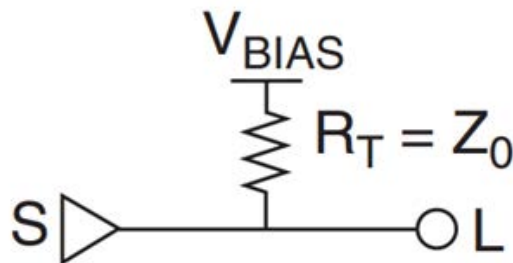
It's a variation of the parallel termination scheme in which the termination resistor is replaced by two resistors whose combined value will be equal to the line impedance. Here, the resistors are placed between V_{CC} and the ground. Hence, the total current drawn from the source is reduced.



Thevenin termination scheme

5.3.3. Active parallel termination

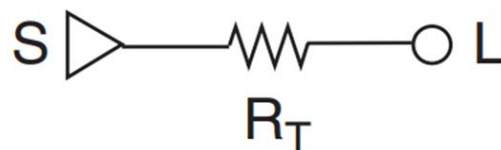
This scheme requires a separate voltage called bias voltage which is connected to R_T . The termination resistor value is kept as same as the line impedance. The arrangement of the bias voltage is such that the output draws current from high and low levels of signal.



Implementing decoupling capacitors to reduce PDN noise

5.3.4 Series termination

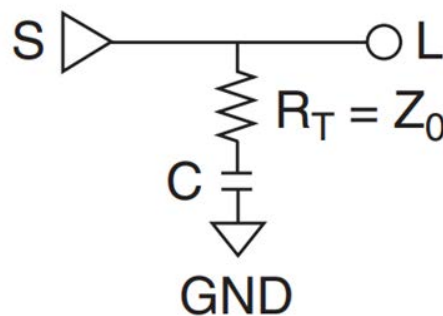
In this scheme, the R_T is placed near to source rather than the load. The value of R_T is made to match the source impedance. A single-value resistor is not applicable for all conditions as the line impedance varies depending on the load distribution. The advantage of this scheme is that it helps to mitigate secondary reflection. Also, only a single component can be implemented at the source contrary to placing multiple components at each load. But, this approach can cause a delay in the signal path due to the increase in the RC time constant.



Series termination

5.3.5 Series-RC parallel termination

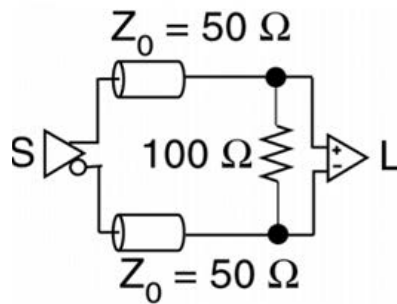
Here, the R_T is placed in series with a capacitor ($>100\text{pF}$) as the terminating impedance. The value of the resistor is kept equal to the line impedance. The capacitor will block the low-frequency components and lets the high-frequency signal components pass through the circuit. Thus, helping to avoid the DC loading effect of R_T on the driver.



Series-RC parallel termination

5.3.6 Differential pair termination

In this scheme, the RT is placed in between the differential signals at the receiving device. Here the termination resistor value should be equal to the differential load impedance (typically 100Ω).



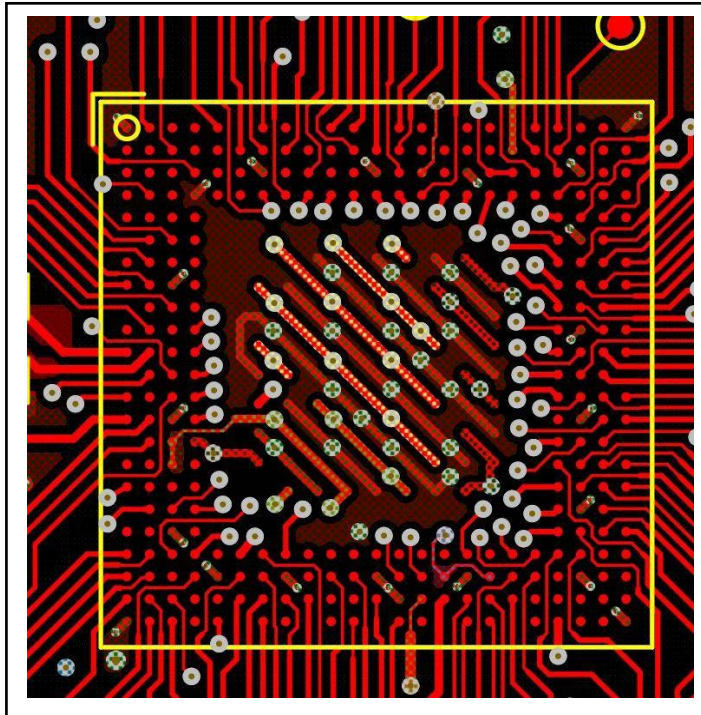
Differential pair termination in a high-speed PCB design

5.4 Component placement and routing strategies

Advanced circuit boards such as high-speed and HDI circuits will have multiple nets embedded in them. Generally, the signal starts propagating at the source of a trace/net, travels through all the connected parts, and terminates at the load. Hence, the pin-to-pin connection between the components should be as short as possible to maintain SI.

In the same concept, it is challenging to route traces for the sensitive components (BGA, TTL chips, precision resistor cards, etc) as they need to be far to avoid interference, but there should be short trace routes connecting them. The following methods can help avoid this issue:

- Provide shielding to sensitive components to mitigate the noise in the signal.
- Implement bypass capacitors close to the power pins of ICs, processors, and memory devices.
- Practice proper BGA fanout strategy, especially with high-density traces. Typically, these traces will be buried using “via-in-pad” technology.



BGA Fanout

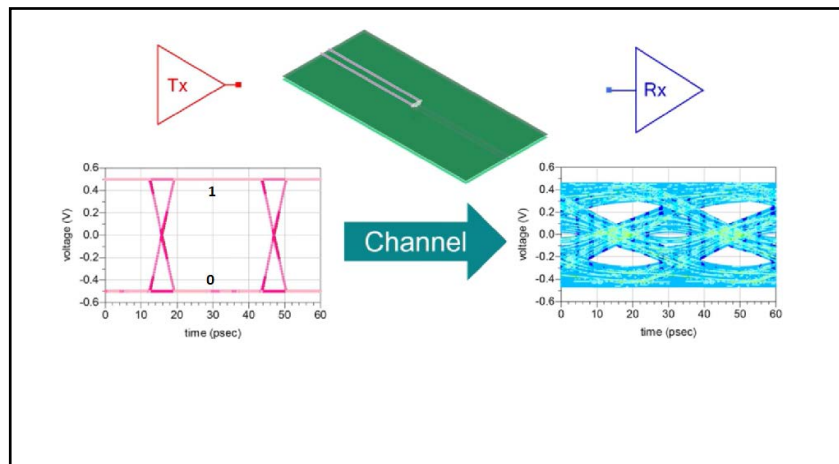
- Due to the high pin density of BGA, a technique called 'necking' is implemented, where the trace width is adjusted at the mid-trace. As you already know this will cause impedance mismatches, mainly in high-frequency circuits.

6. Signal integrity testing

Signal integrity checks will help you identify and resolve the signal integrity issues at the design stage itself. There are in-built signal integrity testing options in the PCB design software. Final signal integrity evaluation of the PCB can also be performed in the following ways:

6.1 Eye diagrams for signal noise measurement

Eye diagrams offer a concise graphical representation of how the channel degrades the signal. An open eye corresponds to minimal signal distortion. Loss and reflections in the system can cause the eye to close, detecting the distortion of the signal.



Eye diagrams at transmitter and receiver

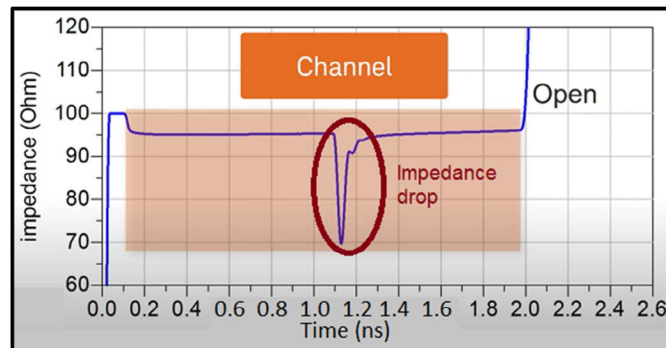
Eye diagrams and bit error rate calculations are performed with an oscilloscope. These measurements are critical for evaluating digital channels to allow quantification of jitter, signal reflections, and losses.

6.2 TDR measurement for impedance discontinuities and signal reflections

Time domain reflectometer (TDR) is a device that analyses reflected waveforms to determine the characteristic impedance in cables/connectors and the controlled impedance of traces.

A TDR measurement is performed in the time domain. It is a measure of the signals reflected from the destination device to the input as a function of time.

In TDR, pulses are fed onto the transmission line on a test coupon. It analyses the changes in the amplitude of the reflected waveform to determine the characteristic impedance. In an ideal case where there are no impedance discontinuities in a transmission line, the injected impulses will be absorbed by the termination component. Hence, proving that there are no reflections in the line. TDR graph represents the change in the impedance.



Impedance measured through TDR

TDR utilizes the velocity factor (VF) to convert the travel time of the reflected pulse into the distance. VF is defined as the ratio of pulse speed on the trace to the speed of light. Hence, it is essential to know the speed of the signal pulse propagating through the trace.

We also have to vary the output pulse to detect the impedance discontinuities. For instance, the high pulse energy is required to locate the discontinuities at the far end of the transmission line, whereas a lower pulse will suffice for the near end.

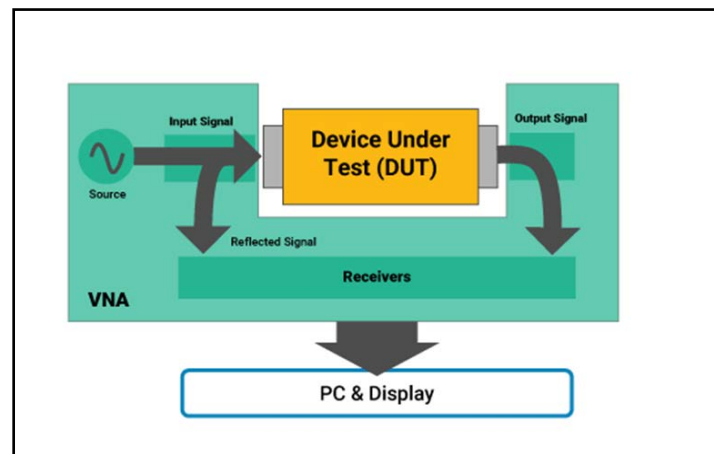
6.3 Coupons for testing impedance

For the bare board impedance test is carried out by placing the correct test point on a test coupon. The test coupon is fabricated on the same panel along the edge of the board. The purpose of this coupon is to accurately depict the stack-up for impedance testing. Instead of creating a separate section, you can also include the coupon in your original design. This is preferred in the prototyping stage for in-circuit testing.

Using test coupons you can also measure controlled impedance, conductor loss, propagation delays, PDN capacitance, and dielectric constant of a substrate.

6.4 Vector network analyzer for S parameters

A VNA comprises a source (with a known stimulus signal) and a receiver that detects the change in the stimulus signal. When the device-under-test (DUT) is connected to VNA, the modification it causes to the stimulus signal is read by the receiver. Typically, VNA measures the power and frequency response of a single component or a network of components (passive and active). It also captures the amplitude and phase of the signal. This is usually used for measuring the S-parameters.

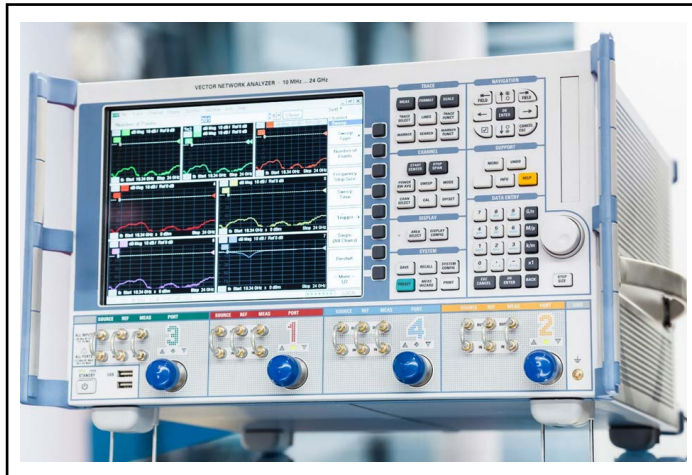


Measurement of signal reflection caused by DUT in a VNA

6.4.1 S-parameters

Scattering parameters depict the energy propagating through an electrical network. It represents a complex network as a simple black box and evaluates the voltage and current at the input and output ports. It captures the network behavior in amplitude and phase vs. frequency. This helps determine the reasons for signal integrity issues such as ground bounce, crosstalk, jitter, and EMI.

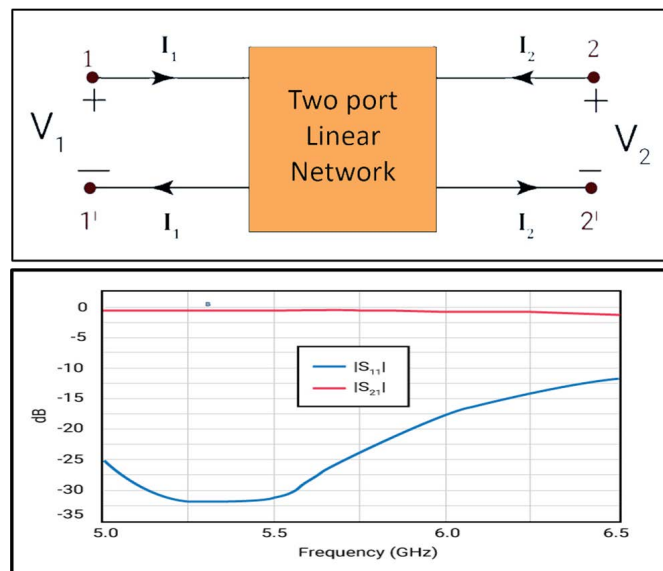
S-parameters have allowed engineers to evaluate complex networks with ease. It also helps to achieve efficient controlled impedance in PCBs.



Vector network analyzer for S-parameters measurement

Single-ended S-parameters: Every combination of input and output are studied in single-ended S-parameters. Here, all the ports have the same source and termination impedances. But there is a difference in the magnitude and phase of the signals between any two ports. The variation is dependent on the characteristics of each path.

For instance, consider a two-port network having a transmission line with a sine wave at port 1. The final result can be obtained after solving the complex ratios of each port combination. The input and output are assigned numbers such as S_{11} (signal going in and coming out of port 1) representing the reflection coefficient and related to return loss. Similarly, S_{21} (signal outgoing from port 2 and incoming to port 1) denotes the transmission coefficient and related to insertion loss.



S-parameter for two port linear network

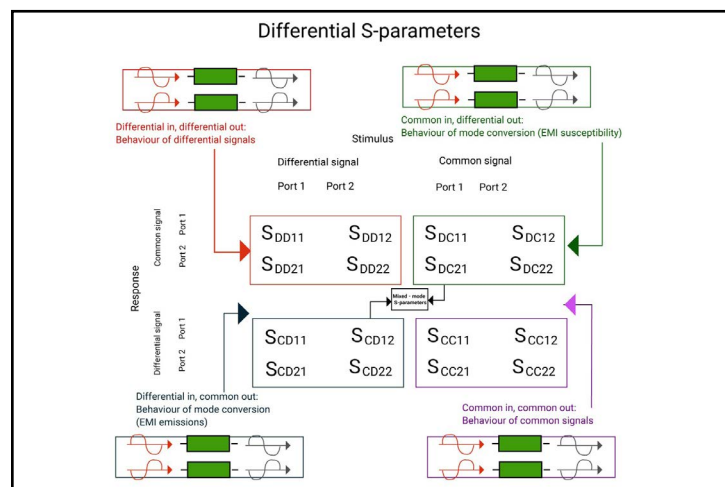
Mixed-mode S-parameters: This mainly deals with ports powered by a differential source. Here, you can only consider two ports. The waveform of the differential ports has two components: differential signal (out-of-phase) and common mode signal (in-phase).

When a signal is injected into a differentials pair, there are four possible outcomes

1	SDD	Differential signal input	Differential signal output
2	SCC	Common signal input	Common signal output
3	SCD	Common signal input	Differential signal output
4	SDC	Differential signal input	Common signal output

Due to these variables, the nomenclature of the mixed mode S parameters is denoted by 'SxxOI'. Where xx can be a differential or common signal. O and I are the output and input ports as represented in the single-ended S parameters.

To keep track of these variables, they are represented in a simplified matrix form as shown in the figure.



- All differential in and out ports (SDDxx): Upper left quadrant
- All common in and out ports (SCCxx): Bottom right quadrant
- Differential and common in and out ports (SDCxx): Top right quadrant
- Common and Differential in and out ports (SCDxx): Bottom left quadrant
- For calculating the mixed mode S-parameters, the single-ended S-parameters are calculated initially. Later, the matrix is created and solved with the following assumptions for the connections:
 - Passive: There is energy loss without energy conversion.
 - Linear: Frequency is constant.
 - Time-invariant: The system is stable without any variation.

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- HDI designs with via-in-pad, blind via, buried via and microvia, via stacking and staggering technology
- Designs with fine pitch BGAs of 0.4 mm and 0.5 mm
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- RoHS compliant design
- Stack-up design and impedance calculation for single-ended, differential, and co-planar-waveguide models

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- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR registered

- ITAR registration letter

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- Rigid PCBs

Exceptional materials compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

Certified minority supplier by NMSDC

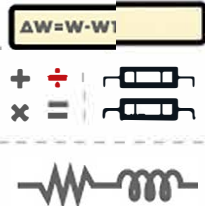
- Minority business certification

The Designer's Toolkit

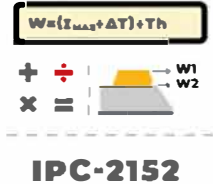
Bandwidth, Rise Time and Critical Length Calculator



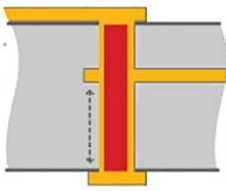
Impedance Calculator



Trace Width, Current and Temp Rise Calculator



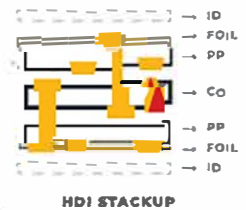
Maximum Via Stub Length Calculator



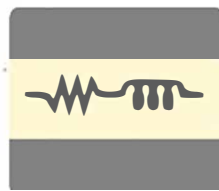
PCB Material Selector



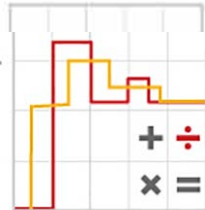
PCB Stackup Designer



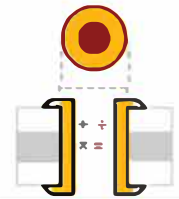
Signal Layer Estimator



Transmission Line Reflection Calculator



Via Current Capacity and Temperature Rise Calculator



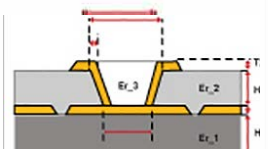
PCB Conductor Spacing and Voltage Calculator



RLC Resonant Frequency & Impedance Calculator



Via Impedance Calculator



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